NLP 2026 Lab 4 Report

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1 Establishing a baseline

To establish a baseline prior to using BERT, we re-adopted the static GloVe embeddings (300-dimensions) as used in lab-2, and combined it with a logistic regression classifier. To handle the sentence pairs (premise and hypothesis), we selected the following sentence-pair combination strategy:

InferSent-style combination (Conneau et al., 2017) - each sentence is mean-pooled into a single vector v_p and v_h. The final feature vector is formed by concatenating their element-wise difference and dot product:

$$f = [v_p \; ; \; v_h \; ; \; |v_p - v_h| \; ; \; v_p \odot v_h]$$

The InferSent approach outperformed the cosine similarity baseline (val=0.6413 vs 0.5954), and was therefore chosen as the baseline for comparison with the proceeding BERT-based methods.

2 Analysis of BERT embeddings

BERT (Bidirectional Encoder Representations of Transformers) is a transformer based Language Model that produces contextual embeddings for tokens in a sentence. The difference to static word embeddings is that BERT generates embeddings that depend on the surrounding context of the token, allowing tokens to have different representations depending on its usage. The embeddings are obtained from BERT's hidden layers.

In this section we analyse two different ways of BERT representations: token-level embeddings and sentence-level embeddings. Token-level embeddings capture contextual information for individual tokens but many Natural Language Processing tasks require a single vector representing an entire sentence. There are several methods to aggregate token information into sentence-level representations.

2.1 Token-level

Token-level embeddings are contextualised representations for a single token. Each token is assigned a numerical high dimension vector capturing its semantic meaning and the relationship to other tokens in the surrounding context using the models self attention mechanism. Thus the same token can have multiple different vector representations depending on the sentence it appears in. Token-level embeddings provide valuable information about the structure and meaning of an input sequence.

2.2 Sentence-level

The task of natural language inference ideally needs a single vector for a whole sentence rather than separate embeddings for each token. There are several ways to obtain such sentence level embeddings.

BERT provides the "[CLS]" token which is added at the beginning of the input sequence. Through the self-attention mechanism of the transformer layers, the representation of the [CLS] token is updated based on the interaction with all the other tokens in the sequence. The resulting final hidden representation is often used as a summary representation for the whole input sequence.

Alternatively, sentence level representations can be obtained by aggregating token embeddings through mean pooling. Instead of relying on only the embedding of the CLS token, the embeddings of all tokens in the sequence are merged into a single vector. This approach captures information throughout the entire sequence as all relevant tokens contribute to the final sentence representation.

3 Methods

The following section describes the methods we used for Natural Language Inference during the lab, starting with the final CLS token representation, followed by a mean pooling approach at the last hidden layer and finally the approach of layer-wise probing analysis for the previously mentioned approaches.

3.1 BERT CLS token pooling

The first approach of finding a sentence level representation was to use the embedding of the [CLS] token. As mentioned previously, the [CLS] token is inserted at the beginning of the input sequence. We used the final hidden representation of this token as a sentence-level embedding of the input sequence.

Each input consisted of a premise-hypothesis pair, which was jointly tokenized using the BERT tokenizer and passed to the BERT model. The resulting CLS token embedding produced by the BERT model was a vector that represents the semantic relationship between the two sentences.

In the next step these sentences were used as input features for a logistic regression classifier, trained on the embeddings extracted from the training set and evaluated on the validation and test set.

For our Natural language inference task, each instance consisted of a premise-hypothesis pair of sentences. Both sentences were tokenised using the BERT tokenizer in order to format them into a single input sequence.

3.2 Sentence-pair representation via mean aggregation of BERT hidden states

As an alternative to the [CLS] token representation of sentences we used sentence-pair representation using pooling over token embeddings. Similar to the previous approach, the input sequences consisting of a premise-hypothesis pair were tokenized and passed to the BERT model. Subsequently the contextual embeddings for all tokens were extracted from the final hidden layer.

In the next step, the embeddings of all tokens were averaged, excluding the padding tokens, ensuring that only valid tokens contributed to the final representation. The resulting mean pooled embeddings were used as the sentence representation vectors. These vectors were passed to a logistic regression classifier, which was validated on the validation and test set.

4 Experiments and Results

4.1 Layer-wise probing analysis

Instead of only evaluating the performance of the representation at the final transformer layer we analysed how the representation at each transformer layer performed. For this purpose, sentence embeddings were extracted from each layer of the BERT model using both the CLS token representation and the mean pooling over token embeddings.

At every layer, embeddings were computed for the training, validation and test sets and used as input for a logistic regression classifier that was then evaluated on the validation set in order to find the layer yielding the best performance.

The results of this experiment can be seen in Figure 1. The layer with the best performance for [CLS]-token-based approach was layer 9 (out of a total of 12 layers) with an accuracy of 0.7322 on the validation set. For the mean pooling approach the best layer was layer 11 with an accuracy of 0.7030. For both approaches the lowest layers show the weakest performance, while the accuracy generally increases in the middle layers before slightly decreasing in the final layers. Starting from layer 4, the CLS embedding approach consistently outperformed the Mean pooling approach.

This experiment shows, that the assumption to take the last layer as the one with the best performance is wrong and earlier layers may produce better results.

4.2 Evaluation: Baselines vs BERT Representations

In the previous subsection, we established the top performing tunings from the two BERT methods (as observed in figure 1). We now place these results in context by comparing them with the baseline model. Figure 2 illustrates the accuracy gain of each BERT model compared to the best performing baseline (GloVe-300d-InferSent). This was done by setting our baseline model as the flat 0 line. The results clearly show that the BERT CLS (L9) gains the greatest accuracy (0.085), implying a slight improvement in classifications capabilities.

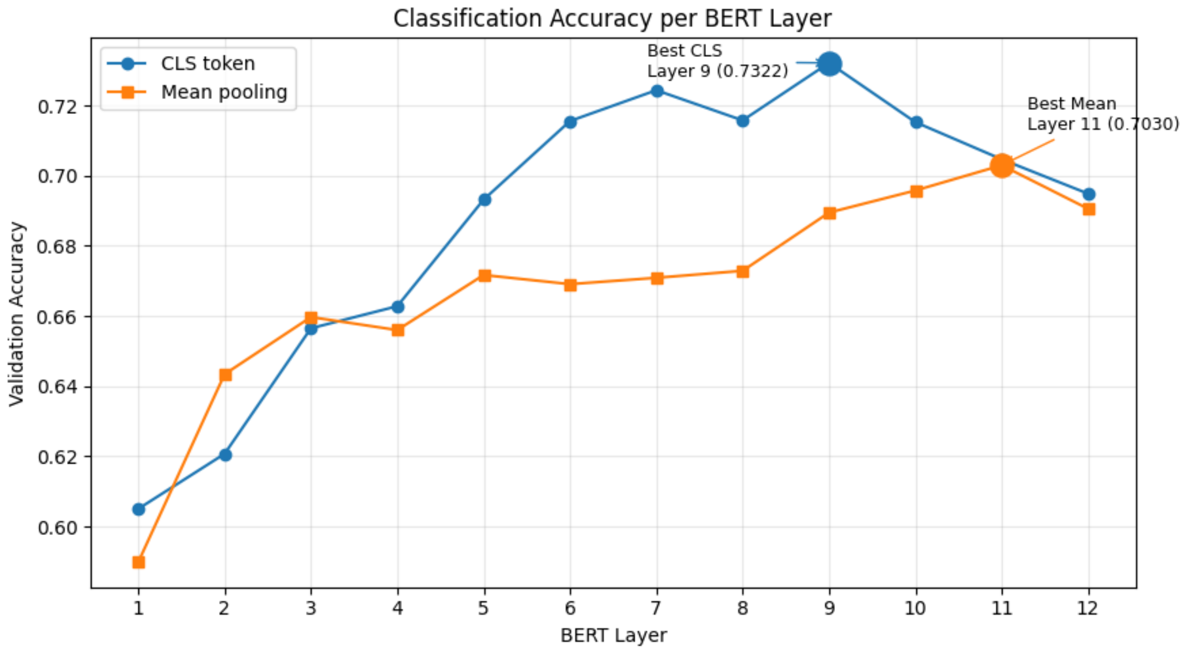

Figure 1: Validation accuracy comparison between BERT (CLS) method and BERT (mean-aggregation-pooling) method at every layer

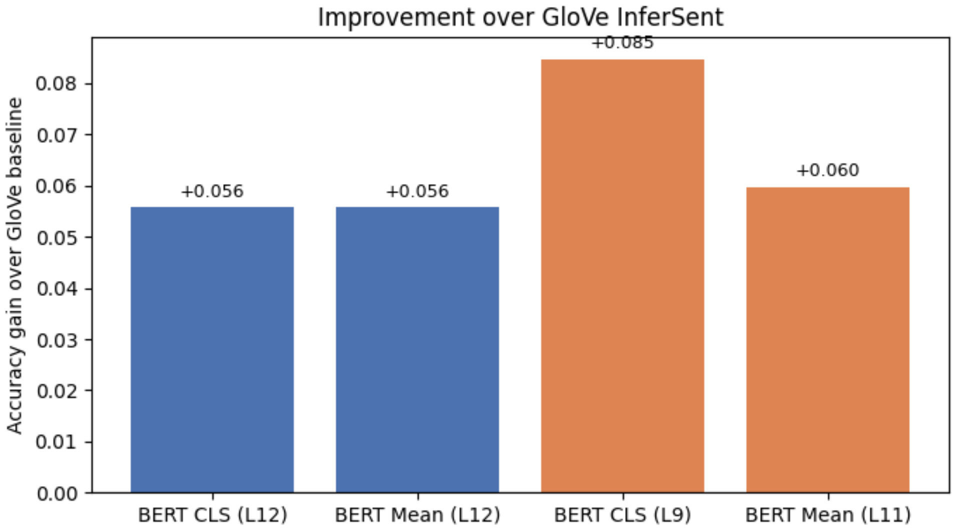

Figure 2: Accuracy gains (from baseline) for each of the BERT method

4.3 Classifications: Baseline vs BERT CLS

To further analyze the performance of the models, we compared the predictions of the BERT CLS model (layer 9) with our GloVe-300d-InferSent baseline model. For both models we created a confusion matrix on the test set to examine how they perform at classifying the labels "Entailment" and "Contradiction". By comparing the confusion matrices we can identify which models makes fewer classification mistakes and if improvement is consistent across both classes or focused on only one of them.

Figure 3 presents the confusion matrices. The BERT CLS model correctly predicted 261 more entailment examples and 306 more contradictions. This is equivalent to making less mistakes by predicting 261 fewer false contradictions and 306 fewer false entailment.

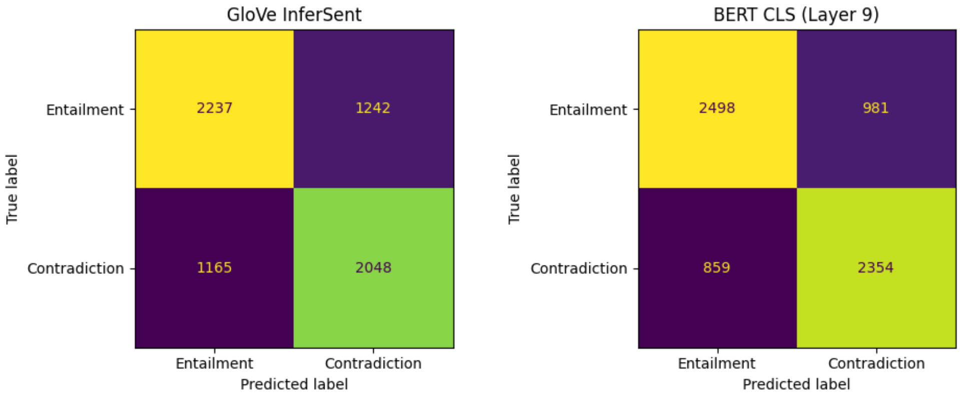

Figure 3: Confusion matrices for GloVe (Baseline) and BERT CLS (9th Layer)

5 Discussion

The results demonstrate a slight, yet clear progression in model performance as we shift from static to contextual embeddings. The original Glove-300d InferSent baseline achieved a validation accuracy of 0.6413 which showed to fundamentally be incapable due to its inability to capture meanings of words that highly depend on the context. Both BERT approaches (CLS and mean-aggregation) outperformed the baseline, with the CLS token representation at layer 9 achieving the highest accuracy of 0.7322 (+0.085 over baseline) (figures 1, 2). Conversely, the mean-aggregation-pooling peaked at layer 11 reaching 0.7030 (+0.062). This is consistent with our initial expectation: BERT's bidirectional self-attention mechanism allows each token's representation to be informed by the full surrounding text, yielding in higher accuracies when performing pairwise semantic comparison between sentences. Figure 1, revealed that the best representations don't necessarily emerge from the final transformer layer. For BERT-CLS, peak performance occurred at layer 9, while mean-aggregation pooling peaked at layer 11. This suggests that topmost layers are over-specialized toward BERT's pre-training objectives, that is masked language modeling and next sentence predictions, rather than general semantic similarity. The confusion matrices in figure 3 support the overall assessment; BERT CLS (L9) model shows improved classifications between Entailment and Contradiction, in comparison with the baseline model. This suggests that contextual representations better capture the logical relationship between sentence pairs, just as expected. Overall, these results confirm that contextual embeddings provide a meaningful improvement. Overall, while both models may struggle with neutral-leaning examples, these results nonetheless confirm that contextual embeddings provide a meaningful improvement for this task. For future improvements, we propose fine-tuning BERT on relevant tasks which would yield substantially higher performance.

References

Alexis Conneau, Douwe Kiela, Holger Schwenk, Loïc Barrault, and Antoine Bordes. 2017. Supervised learning of universal sentence representations from natural language inference data. In *Proceedings of the 2017 conference on empirical methods in natural language processing*, pages 670–680.

A Use of genAI

`Claude Sonnet 4.6` was used as a coding assistant for producing our desired plots. After obtaining our experimental results, it was prompted to help visualize them using `matplotlib`, producing Figures 1, 2, and 3. The generated code was reviewed and adjusted until the visualizations accurately represented our results.